

1. (a) (i) Outline **four** characteristics of object oriented programming language. (4 marks)

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- (ii) Explain the term *abstract data type* as used in OOP. (2 marks)

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- (b) Distinguish between *header file* and *in-built function* as used in C++ programs. (4 marks)

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- (c) Joseph intends use OOP to develop his trade project. Explain **two** benefits he is likely to derive from this decision. (4 marks)

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- (d) Write a C++ program that implements a class named *triangle* with the following properties:

- has data members named base and height;
- a member function named calculate for inputting base and height, determining the area of triangle and outputting the area.

(6 marks)

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2. (a) Distinguish between *portability* and *machine independence* as used in programming. (4 marks)

(b) (i) Explain the following terms as used in OOP:

I. abstraction; (2 marks)

(ii) Outline **three** logical operators used in C++ programs. (3 marks)

**Turn over**

(c) Differentiate between *extension* and *combination* forms of inheritance as used in OOP. (3 marks)

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(d) Interpret the following C++ program segment.

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class student
{
    long int rollno;
private:
    int age;
    char sex;
    float height;
public :
    student();
    void getdata( );
    void disinfo(void);
    int process(int age, int sex);
};
```

**(6 marks)**

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3. (a) Namu intends to design an application module based on object orientation. Explain **three** outcomes of the object design phase. (6 marks)

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- (b) (i) Explain the circumstance under which each of the following concepts are most applicable in C++ programs:

I. comments; (2 marks)

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II. resolution operator. (2 marks)

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- (ii) Explain the term *declaration* as used in programming. (2 marks)

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- (c) Read the following extract and answer the question that follows.

*When ordering new videotapes from a supplier, the store manager creates a purchase order, fills in the date, the supplier's name, address, and enters a list of videotapes to be ordered. The purchase order is added to a permanent list of purchases. When one or more video tapes are received from a supplier, a clerk locates the original purchase order and makes a record of each tape that was received. A record of the videotape is then added to the store's inventory. When all tapes listed on a particular purchase order have been received, the manager sends a payment to the supplier and the purchase order is given a completion date.*

Identity **four** possible *classes* and **four** possible *methods* from the extract. (8 marks)

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4. (a) (i) Outline the general syntax of defining an inline function. (2 marks)

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(ii) Explain the following terms as used in classes:  
I. encapsulation; (2 marks)

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II. instantiation. (2 marks)

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(b) (i) With the aid of an example, describe *explicit type casting* as applied in C++ programs. (3 marks)

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(ii) Distinguish between *member* and *friend* functions as used in C++ programs.

(4 marks)

(c) Write a C++ program that will initialize two objects as follows:

Rectangle 1 L = 10.5 and W = 8

Rectangle 2 L = 7 and W = 3.2

The program should then determine the perimeter of the objects through the use of a *friend* function and output appropriately. Use a constructor. (7 marks)

5. (a) With the aid of a C++ program segment, demonstrate function overloading.

(4 marks)

(b) Distinguish between *object pointer* and *object reference* as used in OOP. (4 marks)

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(c) Explain the circumstance under which each of the following features are used in C++ programs:

(i) friend function; (2 marks)

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(ii) destructor. (2 marks)

1. The first step is to identify the problem or question that needs to be answered. This involves understanding the context and the specific requirements of the task.

(d) Rachael intends to subtract Matrix A [2 3] from Matrix B[7 9]. Write a C++ program that could be used by Rachael to initialize the two objects and use operator overloading to determine the different. The program should output the values of the difference. (8 marks)

[illegible]

6. (a) (i) Outline the stage at which the following objects are destroyed: (1 mark)

1. local object;

II. global object. (1 mark)

(ii) Constructors are essential during object oriented programming. Outline four rules that should be observed when using them. (4 marks)

**Turn over**

**(b) Explain the following terms as used in OOP:**

- (i) pass object by value;

**(2 marks)**

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(ii) overriding.

**(2 marks)**

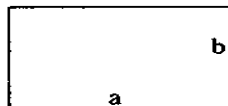
1. The first step in the process of creating a new product is to identify a market need. This involves conducting market research to understand the preferences and behaviors of potential customers. Once a need is identified, the next step is to develop a concept that addresses this need. This concept should be innovative and differentiated from existing products in the market.

2. After developing a concept, the next step is to create a prototype. This allows the company to test the feasibility of the product and gather feedback from potential users. The prototype should be functional and represent the key features of the final product. Based on the feedback received, the company can make necessary adjustments to the design and functionality.

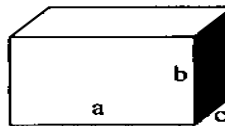
3. Once the prototype is refined, the next step is to conduct a small-scale pilot test. This involves producing a limited quantity of the product and distributing it to a select group of customers. The purpose of the pilot test is to evaluate the product's performance in a real-world setting and gather valuable feedback from actual users. This feedback can be used to make further improvements to the product before launching it on a larger scale.

4. The final step in the process is to launch the product into the market. This involves developing a marketing strategy to create awareness and generate interest among the target audience. The company should also establish distribution channels to ensure the product is readily available to customers. After the launch, the company should continue to monitor the product's performance and gather feedback from customers to make any necessary adjustments or improvements.

(c) Figure 1 shows two objects; object 2 has been derived from object 1. Use it to answer the questions that follow.



### Object 1



## Object 2

**Figure 1**

(i) Write a C++ program that would initialize the values of a and b in object 1 to 7 and 5. The program should allow object 2 to derive the values of a and b but accept the values of c, determine and output the volume of object 2 through the use of a function. (8 marks)

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- (ii) State the *form of inheritance* implemented in the program justifying your answer. (2 marks)

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7. (a) Ben has been instructed by his project supervisor to use OOP. Explain **two** ways he could use to cope with emerging trends in OOP. (4 marks)

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- (b) (i) Define a file as used in OOP. (2 marks)

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- (ii) Outline **three** types of streams used in C++ files. (3 marks)

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(iii) Explain the term *opening a file* as used in C++ programs.

(2 marks)

1. **Identify the problem:** The problem is that the company is not meeting its sales targets.

2. **Analyze the problem:** The company is not meeting its sales targets because it is not effectively marketing its products.

3. **Develop a solution:** The company should develop a new marketing strategy that focuses on reaching its target audience.

4. **Implement the solution:** The company should implement the new marketing strategy by launching a series of targeted advertising campaigns.

5. **Evaluate the solution:** The company should evaluate the effectiveness of the new marketing strategy by tracking sales and customer feedback.

(c) Write a C++ program that will carry out the following:

- defines a class named *polygon* that has data members (*base*, *height*) and a member function named *set* which is used to initialize the values of data members;
- implements a polymorphic function named *area* which determines the area of a triangle and area of a rectangle;
- outputs the area of a triangle and area of a rectangle with base and height as 8cm and 4cm respectively.

(9 marks)

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8. (a) (i) Copy constructors are only applicable during initialization. Outline **three** situations where the constructors could be used during programming. (3 marks)

- (ii) With the aid of an example in C++ programming language, describe a *constructor with arguments*. (3 marks)

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- [illegible]

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